

Eg. Assume 5 women and 3 men are to be ordered in a line.

what's the probability that all the women lined next to each other?

Method 1: we can assume all the women together make one person, together with 3 men there are 4 people, so $4!$ ordering and women can be ordered among themselves in $5!$ ways. Hence

$$|E| = 4! 5!$$

$$P(E) = \frac{|E|}{|S|} = \frac{4! 5!}{8!}$$

method 2: we could order men among themselves in $3!$ way and women among themselves $5!$ and since women must be next to each other we could first position the men

— m_1 — m_2 — m_3 —

and there are 4 places that we could put the

women in so $\binom{4}{1}$. Hence

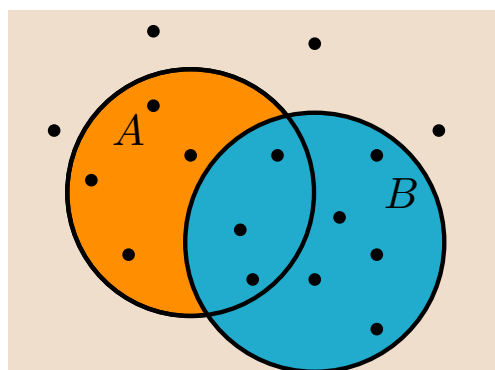
$$|E| = 5! 3! \binom{4}{1} \xrightarrow{\text{if simplified}} 5! 3! 4$$

$$|E| = 5! 4!$$

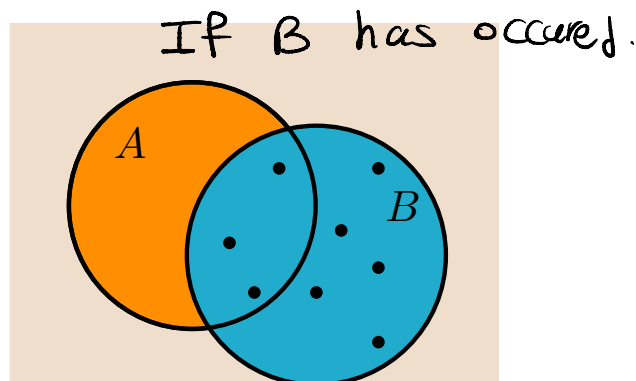
$$\Rightarrow P(E) = \frac{|E|}{18!} = \frac{5! 4!}{8!}$$

Section 2.4 Conditional Probability

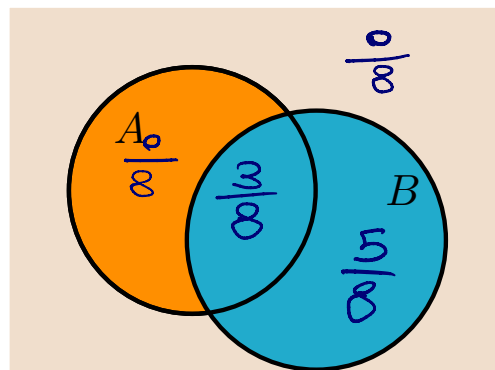
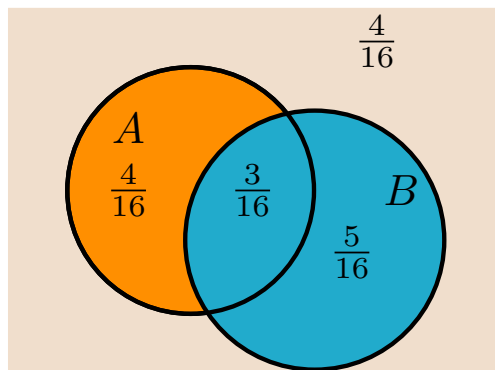
The general Idea: how should we update our model based on the evidence we observe?



$$P(A) = \frac{7}{16}, P(B) = \frac{8}{16}$$



$$P(A|B) = \frac{3}{8}, P(B|B) = 1 = \frac{8}{8}$$



Motivation: Here A is the event whose uncertainty we want to update, and B is the evidence we observe (or want to treat as given). We call $P(A)$ the **prior probability** of A and $P(A|B)$ the **posterior probability** of A (“prior” means before updating based on the evidence, and “posterior” means after updating based on the evidence). It is important to interpret the event appearing after the vertical conditioning bar as the evidence that we have observed or that is being conditioned on: $P(A|B)$ is the **probability of A given B occurred**,

Def 1: Conditional Probability

Let A and B be two events with $P(B) \neq 0$. The conditional probability of A given B , is defined to be

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

or for discrete sample space

$$P(A|B) = \frac{|A \cap B|}{|B|} = \frac{\text{number of outcomes that are both in } A \text{ and } B}{\text{number of outcomes in } B}$$

or for a continuous sample space

$$P(A|B) = \frac{\text{area}(A \text{ intersection with } B)}{\text{area}(B)}$$

Note: Conditional probabilities are often indicated by words/phrases like “given that”, or “if”, or by words implying a subpopulation. Here are some examples of phrases that refer to a conditional probability, along with their translation into probability language.

- “5 percent of those taking drugs test negative.”

Interpretation: “ $P(\text{test negative} \mid \text{take drugs}) = 0.05$.”

- “For each smoker, the probability of dying during the year is 0.05”

Interpretation: “ $P(\text{dying} \mid \text{smoker}) = 0.05$ ”

- “A blood test indicates the presence of a disease 95% of the time the disease is actually present.”

Interpretation: “ $P(\text{test indicates disease} \mid \text{has disease}) = 0.95$ ”

- “In 10% of the spam emails, the phrase free money is used.” Interpretation: “ $P(\text{free money} \mid \text{spam email}) = 0.1$ ”

Eg 1. The probability that a randomly selected student at UMD owns a Scooter is 0.55, the probability that a student owns a car is 0.30, and the probability that a student owns both is 0.10.

- (a) What is the probability that a student owns a scooter, given that he/she owns a car?
- (b) Suppose a student does not have a scooter. What is the probability that he/she has a car?

S = A student has a scooter

C = A student has a car

$$P(S) = 0.55$$

$$P(C) = 0.3$$

$$P(S \cap C) = 0.1$$

$$(a) \quad P(S|C) = \frac{P(S \cap C)}{P(C)} = \frac{0.1}{0.3} = \frac{1}{3}$$

$$(b) \quad P(C|S') = \frac{P(C \cap S')}{P(S')} = \frac{P(C) - P(C \cap S)}{1 - P(S)} = \frac{0.3 - 0.1}{1 - 0.55} = \frac{0.2}{0.45} = \frac{20}{45}$$

Eg 2. You roll two dice. What is the probability that the sum is ≥ 7 given the first die is ≥ 3 .

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(6,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

$A =$ The sum of two dice is $\geq 7 = \bullet \quad |A| = 21$

$B =$ The first die ≥ 3 . $|B| = 24$

$$p(A|B) = \frac{|A \cap B|}{|B|} = \frac{18}{24} = \frac{3}{4}$$

Eg 3. You roll two dice. What is the probability that the second die is 1, given that the sum of two die is 7 or 8.

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(6,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

A = Sum of the two dice is 7 or 8 $|A| = 11$

B = The second die is 1.

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{|A \cap B|}{|A|} = \frac{1}{11}$$

Def 2: Multiplication rule

$$P(A \cap B) = P(A) P(B|A) = P(B) P(A|B)$$

$$P(A \cap B \cap C) = P(A) P(B|A) P(C|A \cap B)$$

Eg 4. Suppose you deal two cards (in the usual way without replacement). What is $P(\text{two hearts in a row})$.

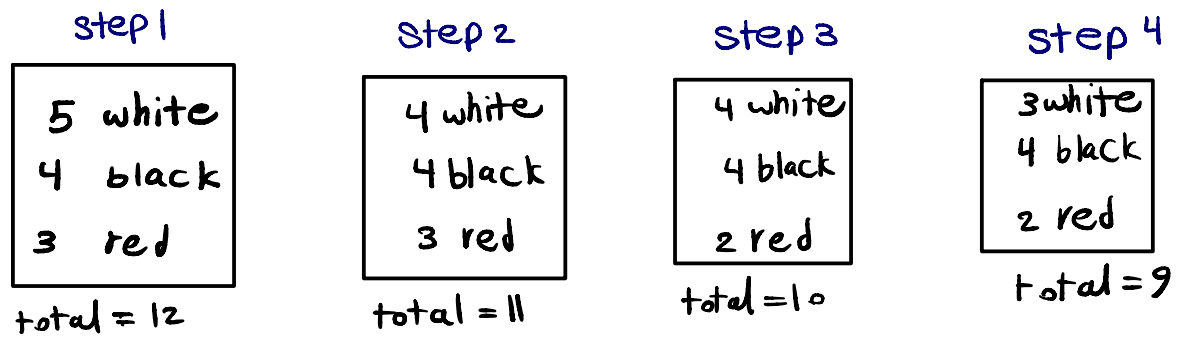
$A = \text{First card is a heart}$

$B = \text{Second card is a heart}$

$$P(A \cap B) = P(A) \cdot P(B|A)$$

$$P(A \cap B) = \frac{13}{52} \cdot \frac{12}{51}$$

Eg 5. An urn contains 5 white balls, 4 black balls and 3 red balls. Four balls are drawn sequentially without replacement. Find $P(WRWB)$.



$$P(W_1 R_2 W_3 B_4) = P(W_1) \cdot P(R_2 | W_1) \cdot P(W_3 | W_1 \cap R_2) \cdot P(B_4 | W_1 \cap R_2 \cap W_3)$$

$$P(WRWB) = \left(\frac{5}{12}\right) \cdot \left(\frac{3}{11}\right) \cdot \left(\frac{4}{10}\right) \cdot \left(\frac{4}{9}\right)$$

Def 3: Partition

A collection of sets $A_1, A_2, \dots, A_n$ is said to partition the sample space if the sets

(i) are mutually disjoint; $A_i \cap A_j = \phi$ for $i \neq j$ and

(ii) have as union the entire sample space; $\bigcup_{i=1}^n A_i = S$

A simple example of a partition is given by a set A , together with its complement A' .

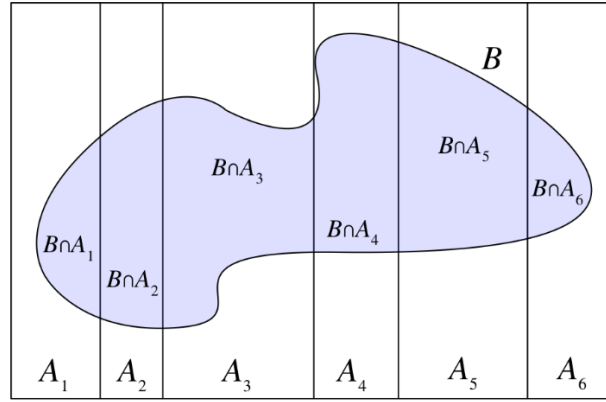


Figure 1: In this picture $A_1, A_2, \dots, A_6$ are a partition for the sample space. The shaded purple area is the event B .

Def 4: Law of Total Probability (LOTP)

Let $A_1, A_2, \dots, A_n$ be a partition for the sample space with $P(A_i) > 0$ for all i . Then, for any event B ,

$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i)$$

or

$$P(B) = P(B|A_1)P(A_1) + P(B|A_2)P(A_2) + \dots + P(B|A_n)P(A_n)$$

Eg 6. We have an infinite collection of biased coins, indexed by the positive integers. Coin n has probability $\frac{1}{2^n}$ of being selected. A flip of coin n results in head with probability $\frac{1}{4^n}$. We select a coin and flip it. What is the probability that the result is head?

The probability that the result is head is:

$$P(\text{coin } n) = \frac{1}{2^n} \quad P(H | \text{coin } n) = \frac{1}{4^n}$$

$$P(H) = \sum_{n=1}^{\infty} P(\text{coin } n) P(H | \text{coin } n)$$

$$P(H) = \sum_{n=1}^{\infty} \frac{1}{2^n} \cdot \frac{1}{4^n} = \sum_{n=1}^{\infty} \left(\frac{1}{8}\right)^n$$

$$P(H) = \frac{\frac{1}{8}}{1 - \frac{1}{8}} = \frac{\frac{1}{8}}{\frac{7}{8}} = \frac{1}{7}$$

Def 5: Bay's rule

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

Eg 7. You have one fair coin, and one biased coin which lands Heads with probability $3/4$. You pick one of the coins at random and flip it three times. It lands Heads all three times. Given this information, what is the probability that the coin you picked is the fair one?

fair coin $P(H) = P(T) = \frac{1}{2}$

Biased coin $P(H) = \frac{3}{4}$ $P(T) = \frac{1}{4}$

fair coin $= C_f$

Biased coin $= C_b$

$P(C_f) = P(C_b) = \frac{1}{2}$

$$P(C_f | HHH) = \frac{P(HHH | C_f) P(C_f)}{P(HHH)}$$

$$P(HHH) = P(HHH | C_f) P(C_f) + P(HHH | C_b) P(C_b)$$

$$P(HHH) = \left(\frac{1}{2}\right)^3 \cdot \frac{1}{2} + \left(\frac{3}{4}\right)^3 \cdot \frac{1}{2}$$

$$P(HHH) = \frac{1}{16} + \frac{27}{128} = \frac{8+27}{128} = \frac{35}{128}$$

$$P(C_f | HHH) = \frac{P(HHH | C_f) P(C_f)}{P(HHH)} = \frac{\frac{1}{16}}{\frac{35}{128}} = \frac{1}{16} \cdot \frac{128}{35} = \frac{8}{35}$$

Eg 8. A spam filter is designed by looking at commonly occurring phrases in spam. Suppose that 80% of email is spam. In 10% of the spam emails, the phrase “free money” is used, whereas this phrase is only used in 1% of non-spam emails. A new email has just arrived, which does mention “free money”. What is the probability that it is spam?

$$p(\text{spam}) = 0.8$$

$$p(\text{“free money”} \mid \text{spam}) = 0.1$$

$$p(\text{“free money”} \mid \text{nospam}) = 0.01$$

$$p(\text{spam} \mid \text{“free money”}) = \frac{p(\text{“free money”} \mid \text{spam}) p(\text{spam})}{p(\text{“free money”})}$$

$$p(\text{“free money”}) = p(\text{“free money”} \mid \text{spam}) p(\text{spam}) + p(\text{“free money”} \mid \text{nospam}) p(\text{nospam})$$

$$p(\text{“free money”}) = 0.1 \cdot 0.8 + 0.01 \cdot 0.2 = 0.08 + 0.002 = 0.082$$

$$p(\text{spam} \mid \text{“free money”}) = \frac{0.08}{0.082} = \frac{80}{82}$$

Def 6 and the example after that is only for your knowledge, and we will not use it in the quiz or home work.

Def 6

Fix B with $P(B) \neq 0$.

$P(\cdot|B)$, ($P(A|B)$ as a function of A) is a probability function.

That means it satisfies the axioms of probability For example

$$P(A_1 \cup A_2|B) = P(A_1|B) + P(A_2|B) - P(A_1 \cap A_2|B)$$

$$P(A'|B) = 1 - P(A|B)$$

$P(A|\cdot)$, ($P(A|B)$ as a function of B) is not a probability function

That means it does not satisfy the axioms and theorems. For example

$$P(A|B') \neq 1 - P(A|B)$$

Eg 9. Events A and B are given.

$$P(A) = 0.4, \quad P(B) = 0.8, \quad P(A \cap B) = 0.2, \quad P(B') = 0.2$$

$$P(A|B') = \frac{P(A \cap B')}{P(B')} = \frac{P(A) - P(A \cap B)}{P(B')} = \frac{0.4 - 0.2}{0.2} = 1$$

$$1 - P(A|B) = 1 - \frac{0.2}{0.8} = 1 - \frac{2}{8} = 1 - \frac{1}{4} = \frac{3}{4}$$

$$\Rightarrow 1 - P(A|B) \neq P(A|B')$$

